In [1]:
```python
import pandas as pd
import numpy as np
```

In [2]:
```python
df=pd.read_csv('iris.csv')
df.describe() # Print description of DataFrame
```

Out[2]:

	sepal.length	sepal.width	petal.length	petal.width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.057333	3.758000	1.199333
std	0.828066	0.435866	1.765298	0.762238
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

In [3]:
```python
print("First 5 values:\n", df.head())
print ("Last 5 values:\n", df.tail())
```

```
First 5 values:
    sepal.length  sepal.width  petal.length  petal.width variety
0           5.1          3.5           1.4          0.2  Setosa
1           4.9          3.0           1.4          0.2  Setosa
2           4.7          3.2           1.3          0.2  Setosa
3           4.6          3.1           1.5          0.2  Setosa
4           5.0          3.6           1.4          0.2  Setosa
Last 5 values:
      sepal.length  sepal.width  petal.length  petal.width    variety
145           6.7          3.0           5.2          2.3  Virginica
146           6.3          2.5           5.0          1.9  Virginica
147           6.5          3.0           5.2          2.0  Virginica
148           6.2          3.4           5.4          2.3  Virginica
149           5.9          3.0           5.1          1.8  Virginica
```

In [4]:
```python
df.duplicated()
```

Out[4]:
```
0      False
1      False
2      False
3      False
4      False
       ...
145    False
146    False
147    False
148    False
149    False
Length: 150, dtype: bool
```

In [5]:
```
df.isnull()
```

Out[5]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	False	False	False	False	False
1	False	False	False	False	False
2	False	False	False	False	False
3	False	False	False	False	False
4	False	False	False	False	False
...	...	...	...	...	...
145	False	False	False	False	False
146	False	False	False	False	False
147	False	False	False	False	False
148	False	False	False	False	False
149	False	False	False	False	False

150 rows × 5 columns

In [6]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [7]:
```
df.shape
```

Out[7]:
```
(150, 5)
```

In [8]:
```
df["sepal.length"].isnull()
```

Out[8]:
```
0      False
1      False
2      False
3      False
4      False
       ...
145    False
146    False
147    False
148    False
149    False
Name: sepal.length, Length: 150, dtype: bool
```

In [9]:
```
y = df.drop(["petal.length"], axis=1) # axis=1 column. For row, axis=0
print(y)
```

```
     sepal.length  sepal.width  petal.width    variety
0             5.1          3.5          0.2     Setosa
1             4.9          3.0          0.2     Setosa
2             4.7          3.2          0.2     Setosa
3             4.6          3.1          0.2     Setosa
4             5.0          3.6          0.2     Setosa
..            ...          ...          ...        ...
145           6.7          3.0          2.3  Virginica
146           6.3          2.5          1.9  Virginica
147           6.5          3.0          2.0  Virginica
148           6.2          3.4          2.3  Virginica
149           5.9          3.0          1.8  Virginica

[150 rows x 4 columns]
```

In [10]:
```
df['variety'].replace(['Setosa', 'Virginica'], [0,1], inplace=True)
print(df)
```

```
     sepal.length  sepal.width  petal.length  petal.width variety
0             5.1          3.5           1.4          0.2       0
1             4.9          3.0           1.4          0.2       0
2             4.7          3.2           1.3          0.2       0
3             4.6          3.1           1.5          0.2       0
4             5.0          3.6           1.4          0.2       0
..            ...          ...           ...          ...     ...
145           6.7          3.0           5.2          2.3       1
146           6.3          2.5           5.0          1.9       1
147           6.5          3.0           5.2          2.0       1
148           6.2          3.4           5.4          2.3       1
149           5.9          3.0           5.1          1.8       1

[150 rows x 5 columns]
```

```
C:\Users\Hp\AppData\Local\Temp\ipykernel_22888\3783490149.py:1: FutureWarning: A val
ue is trying to be set on a copy of a DataFrame or Series through chained assignment
using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because
the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method
({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform
the operation inplace on the original object.


  df['variety'].replace(['Setosa', 'Virginica'], [0,1], inplace=True)
```

In [11]:
```
df.isnull().sum()
```

Out[11]:
```
sepal.length    0
sepal.width     0
petal.length    0
petal.width     0
variety         0
dtype: int64
```

In [ ]: